J. Geomag. Geoelectr., **32**, Suppl. III, SIII 77–SIII 98, 1980

Magnetism of the Mid-Ordovician Tramore Volcanics, SE Ireland, and the Question of a Wide Proto-Atlantic Ocean

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(Received June 25, 1980)

Useful evidence regarding a Proto-Atlantic (Iapetus) ocean may be obtained from paleomagnetic comparisons across sutures in presumed hybrid blocks such as Newfoundland, Britain and Ireland. Usually this involves Lower Paleozoic results that may be further compared with data from larger cratonic units, e.g. Europe and North America. Towards these ends the first paleomagnetic results are reported for an igneous suite comprising andesite sills and flows, pillow lavas, tuffs, mafic intrusives and rhyolite collected from a 20-km wide coastal section near Tramore, Co. Waterford, Eire. From associated fossil evidence the rocks range in age from Lower Caradoc to somewhat earlier (Llanvirn?) Ordovician time, or a mean age of about 460 Myr.

Although at 12 out of the 28 sites studied the remanence of the samples after demagnetization was too weak to measure, or was randomly or anomalously directed, the mean vectors of the remaining 16 sites (76 samples) after AF or thermal treatment and geological tilt correction became well grouped in sharp misalignment with the present Earth's field direction. At most sites the vector groupings of the samples improved upon demagnetization. These results indicate long-term stability of the remanence. At the 6 oldest stable sites this remanence is of reversed polarity and is nearly antiparallel to the (normal) mean direction of the 10 youngest sites, giving an overall north pole at 11°S, 162°W (dp, $dm = 10°$, 13°; $N = 16$ sites). Confidence that the magnetization reflects a mid-Ordovician field direction is based on 1) a strongly positive fold test demonstrating that this magnetization pre-dates the Caledonian folding, 2) the presence of both polarities, and 3) the other long-term stability evidence cited.

The Tramore pole significantly diverges from most published mid- to late Ordovician paleomagnetic poles for the British Isles, including the Mweelrea ignimbrites in western Ireland. The data support a 30° rotation of the British Isles relative to the geomagnetic field during the Ordovician. The 30° angular difference between the Tramore and Mweelrea poles can be reconciled only through relative rotation between the two sampling localities in a sense that places them on opposite margins of a mid-Ordovician Proto-Atlantic ocean. Application of these Irish paleomagnetic results to the plate tectonic model of Phillips, Stillman and Murphy yields a unique rotation pivot for post mid-Ordovician closure of a Proto-Atlantic ocean $3{,}300 \pm 2{,}200$ km wide. Ordovician paleomagnetic comparisons across the North Atlantic suggest that a total Proto-Atlantic reconstruction is premature.

1. Introduction

Three islands, Britain, Ireland and Newfoundland, play key roles in the proposal of a Proto-Atlantic ocean (WILSON, 1966) where each island is considered to be a hybrid block welded from crustal segments that lay on opposite coasts in Lower Paleozoic times. A paleomagnetic scheme for testing this proposal (DEUTSCH, 1969) requires that, on any one island, synchronous rocks exposed on opposite sides of the presumed suture will show stable but divergent directions of magnetization. In Ireland, the first pre-Tertiary formations investigated magnetically are the Mweelrea ignimbrites of Llanvirn age, where three studies (DEUTSCH and SOMAYAJULU, 1970; MURTHY and DEUTSCH, 1971; MORRIS *et al.*, 1973) have established a reliable direction of magnetization. The mid-Ordovician age of these rocks is relevant since geological evidence from the Caledonides of the British Isles (DEWEY, 1969) suggests that the ignimbrites are intermediate in age between the (Lower Ordovician) time of maximum width of the presumed Proto-Atlantic and that of its closure at or after the end of the Ordovician.

The Mweelrea ignimbrites are located northwest of the proposed plate suture (PHILLIPS *et al.*, 1976). Comparisons with paleomagnetic results from Ireland southeast of the suture (P. MORRIS and ROBINSON, 1971; W. A. MORRIS, 1976) have so far proved inconclusive because of discordant magnetization directions among the latter results which the authors variously attributed to remagnetization, insufficient sampling or unknown causes. A relevant though less direct paleomagnetic comparison involving Ireland is with Britain, where several reliable mid- to late Ordovician results are available (summarized in IRVING *et al.*, 1976; FALLER *et al.*, 1977; PIPER, 1979). Most of the British sites are south of the presumed plate suture and yield in general very similar pole positions from which, moreover, one pole located north of the "suture" (Aberdeenshire gabbros; SALLOMY and PIPER, 1973) departs only marginally. The combined British pole, however, is 30 degrees away from the Mweelrea ignimbrite pole. BRIDEN *et al.* (1973) and MORRIS (1976) concluded that, within paleomagnetic error, the pole during the mid- to late Ordovician remained essentially stationary relative to the whole British Isles and any closure across the Caledonides must have been small ($1{,}000 \pm 800$ km). They proposed that the discordant magnetization direction of the ignimbrites was caused locally by a 30-degree net horizontal rotation of the South Mayo trough and that this invalidates the original explanation (DEUTSCH, 1969) of the Mweelrea result in terms of a large Proto-Atlantic ocean.

In this paper, paleomagnetic results are presented for the first time from the widely exposed igneous suite near Tramore, County Waterford, southeast of the suture in Ireland. From fossil dating, the rocks are of mid-Ordovician age comparable to the Mweelrea ignimbrite age. The results, when compared with other paleomagnetic data for the British Isles, call for re-examination of the question of a wide Proto-Atlantic in this section of the Caledonite belt. By combining the paleomagnetic evidence from the two sides of the Irish suture with a plate tectonic model (PHILLIPS *et al.*, 1976; PHILLIPS, 1980) it is attempted here to estimate quantitatively the width of the inferred Proto-Atlantic ocean in mid-Ordovician times.

2. Geology and Sampling

A 25-km coastal strip of largely volcanic rocks west of Tramore, County Waterford, forms part of a wider belt of Cambrian to Silurian volcanism in south-eastern Ireland which peaked in the Upper Ordovician. The rocks are largely submarine and are considered (PHILLIPS *et al.*, 1976; STILLMAN, 1978; STILLMAN and WILLIAMS, 1978) to have been erupted at a continental plate margin related to a SE-dipping subduction zone and developing arc-trench system that formed part of a Lower Paleozoic Proto-Atlantic ocean; such a subduction system was postulated earlier for the Lake District and Wales (FITTON and HUGHES, 1970). STILLMAN *et al.* (1974), SCHIENER (1974), and STILLMAN (1978) have described the geology west of Tramore, where the volcanism is believed to have begun in Llandeilo time or somewhat earlier and was interrupted by deposition of the Tramore limestone, a fossil marker horizon of Llandeilo age (CARLISLE, 1979). The volcanism in this area ended in the Lower Caradoc (CARLISLE, 1979) and the rocks were deformed during the Caledonian orogeny. The volcanic rocks of the Waterford/Tramore region are, in decreasing order of abundance, rhyolite, basalt, andesite and dacite, the acidic rocks being predominantly pyroclastic or intrusive (STILLMAN and WILLIAMS, 1978). Alteration seems to be due mainly to weathering during or shortly after emplacement, or hydrothermal metasomism; rocks affected by a superimposed burial metamorphism commonly are of lower than greenshist facies grade (STILLMAN and WILLIAMS, 1978).

Paleomagnetic collections were made in 1971 and 1976, by the author with logistical assistance from the Geological Survey of Ireland and the Dublin Institute of Advanced

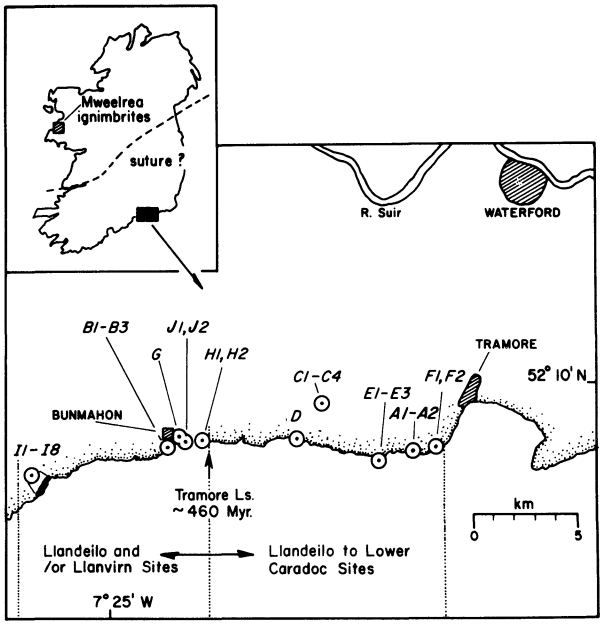


Fig. 1. Tramore area, location of sampling sites.

Table 1. Tramore area: site-mean remanence directions and Fisher statistics.[1]

Site No.[2]	Rock type	Natural remanence (NRM)					After demagnetization				After tilt correction		
		D (°)	I (°)	N	k^3	Treatment[4]	D (°)	I(°)	N	k^3	D (°)	I (°)	α_{95} (°)
A1	Mafic dike or sill	240	0	5	2.1*	40 mT	206	−39	5	4.1	305	−62	42
B1	Andesite (flow?)	307	−15	5	21	40 mT	309	−32	5	14	308	−60	21
B2	As B1	314	+01	4	16	40 mT	315	−28	4	22	317	−56	20
B3	As B1	359	−10	4	<2*	40 mT	353	−43	4	3.2*	—	—	—
C1–C3	Samples from three acid tuff bands[5]	204	+03	5	7.0	—	—	—	—	—	—	—	—
D	Banded rhyolite	142	−64	3	2.8*	40 mT	152	−53	3	2.2*	—	—	—
E1	Acid-intermediate tuff	313	−16	3	109	40 mT	328	−34	3	107	356	−74	12
E2	As E1	261	+06	3	14	40 mT	309	−24	3	33	301	−66	22
E3	Mafic dike	312	+25	3	2.0*	40 mT	331	−54	3	5.9	331	−54	57
F1	Andesite pillows	219	+20	6	6.5	40 mT	237	−13	6	4.7	(230)	(−22)	(34)[10]
F2	As F1	310	−31	5	6.8	40 mT	319	−43	5	70	333	−64	9.2
G	Andesite pillows	215	+55	6	2.7	500 C[7]	294	+32	6	<2*	—	—	—
H1	Andesite flow	289	0	9	9.3	40 mT	273	−37	9	5.3	264	−53	25
H2	As H1	333	+62	5	5.3	40 mT	326	+60	5	93	(317)	(+43)	(8.0)[10]
I1	Andesite sill[6]	263	+58	6	5.1	200 C	241	+49	6	7.1	(1)	(+25)	(27)[10]
I2	As I1	191	−15	5	2.0*	200 C	200	−16	4[9]	12	167	+85	27
I3	As I1	211	+17	5	2.0*	200 C	177	−16	4[9]	4.4	136	+64	49
I4	As I1	188	+49	6	<2*	200 C[8]	194	+10	6	<2*	—	—	—
I5	As I1	218	+11	6	4.7	200 C	208	−08	5[9]	26	183	+71	15
I6	As I1	191	−20	5	2.3*	300 C	190	−18	5	6.7	178	+60	32
I7	As I1	185	−06	5	15	300 C	189	−26	4[9]	37	183	+52	15
I8	As I1	193	+36	5	3.2	300 C	206	+28	5	3.9	103	+53	38
J1	Andesitic tuff	318	−41	6	32	500 C	322	−46	6	116	342	−41	6.2
J2	Andesite sill[6]	314	−08	6	<2*	500 C	315	−41	6	39	333	−39	12

[1] D = declination, deg., E of N; I = inclination, deg., positive downward; N = number of samples averaged (1 specimen per sample); k = precision parameter; α_{95} = radius of 95% confidence circle.

[2] Two tuff sites, A2 and C4 (4 samples each) are omitted as the samples were too weak to measure.

[3] An asterisk (*) after a k-value denotes that within-site vector grouping is random at 95% probability level (IRVING, 1964, p. 63). Sites with random grouping after demagnetization were excluded from final averaging.

[4] Values are for stepwise AF or thermal treatment (see text).

[5] NRM values shown are based on 5 samples from undifferentiated sites C1, C2, C3 measured on the cryogenic magnetometer at Ohio State University. All ten samples from these sites were too weak to measure on the spinner.

[6] Sills probably formed by syn-depositional intrusion into unconsolidated host rock (see text).

[7] Directions are random also at 300 and 600 C.

[8] Directions are random also at 300 and 500 C.

[9] One greatly divergent sample ($>65°$ from mean) was omitted from averaging.

[10] Tilt-corrected sites I1 and F1 were omitted as their directions diverge sharply ($>65°$) from the respective mean directions of site groupings (D, $I=140°$, $+76°$ for 7 I-sites incl. I1; D, $I=309°$, $-60°$ for 11 other sites incl. F1). Discordant site H2 was omitted because of uncertainty in geologic structure (W. Stillman, personal communication).

Studies and with guidance in the selection of sampling sites from Dr. W. Stillman of Trinity College and Dr. E. Schiener of University College, Dublin. A total of 134 oriented drill cores and hand samples were collected from 28 sites (Fig. 1), comprising andesite sills and flows, pillow lava, tuffs, mafic intrusives and some rhyolite (Table 1). The 12 easterly sites (A1, A2, C1–4, D, E1–3, F1, F2) overlie the Tramore limestone and are presumably Lower Caradoc to Llandeilo in age, or about 445–462 Myr on the Elsevier time scale (VAN EYSINGA, 1975). The 16 westerly sites (B1–3, G, H1, H2, I1–8, J1, J2) lie below the limestone and are thus Llandeilo or older (>450 Myr), the oldest sites (I1–8) being probably Llanvirn (M. Boland, personal communication), i.e. 473–462 Myr. Thus the widest probable age range at the 28 sites is 473–445 Myr, with a mean value of about 460 Myr.

The samples were oriented with a Brunton compass which was checked against a solar compass whenever possible, i.e. at most sites. No systematic differences between magnetic and solar readings were found and there seems to be no reason to suspect the Brunton readings taken on other outcrops on sun-less days. Dip and strike values accurate to 2° or better were obtained at nearly every site, usually as the mean of several determinations. At sites other than the I-sites, the dips are typically 20–45°. As there were no indicators, such as plunging fold axes, of serious secondary deformation at any site, it was considered safe to reconstruct the original bedding positions by back-tilting about the strike.

Sites I1 to I8 comprise eight andesite sills, each 10–60 m thick, separated by thick tuff, slate or shale layers. The strata have been tilted by about 90° into their present near-vertical attitudes. These andesites, and probably those at site J2, are believed to have been injected horizontally into the unconsolidated wet sediment close to the sediment-water interface (SCHIENER, 1974). Since sites I1 and I8 (top and bottom sills) are separated by more than 600 m, it is plausible that geologically significant time intervals separated the injection of the different sills and therefore, that secular variation components in the primary remanent magnetization of the sills tend to average out between different sills. The time span represented by the whole Tramore sampling area is almost certainly adequate to have cancelled Earth's field asymmetries of 10^3–10^5 years' duration.

3. Remanent Magnetization

At the laboratory the samples were cut into two or more cylindrical specimens of 2.2 cm diameter and 1.8 cm mean height. The following apparatus was used: (i, ii) Two spinner magnetometers (Princeton Applied Research and Schonstedt); (iii) Alternating-field (AF) demagnetizer with 3-axis tumbler (PEARCE, 1967), for $\leqq$100 mT peak field; (iv) Thermal demagnetizer attaining 800 °C; the original unit (DEUTSCH and SOMAYAJULU, 1970) was improved by addition of large Parry coils and a feedback system confining the estimated maximum field at any specimen to 5 nT. Supplementary equipment included (v) a high-field ($\leqq$350 mT peak) AF demagnetizer; (vi) a DC electromagnet for producing high-field isothermal remanence (IRM); and (vii) a bridge for measuring temperature dependence of susceptibility (PÄTZOLD, 1972) between −196 C and 800 C in a field of 0.03 mT. All measurements were made in air.

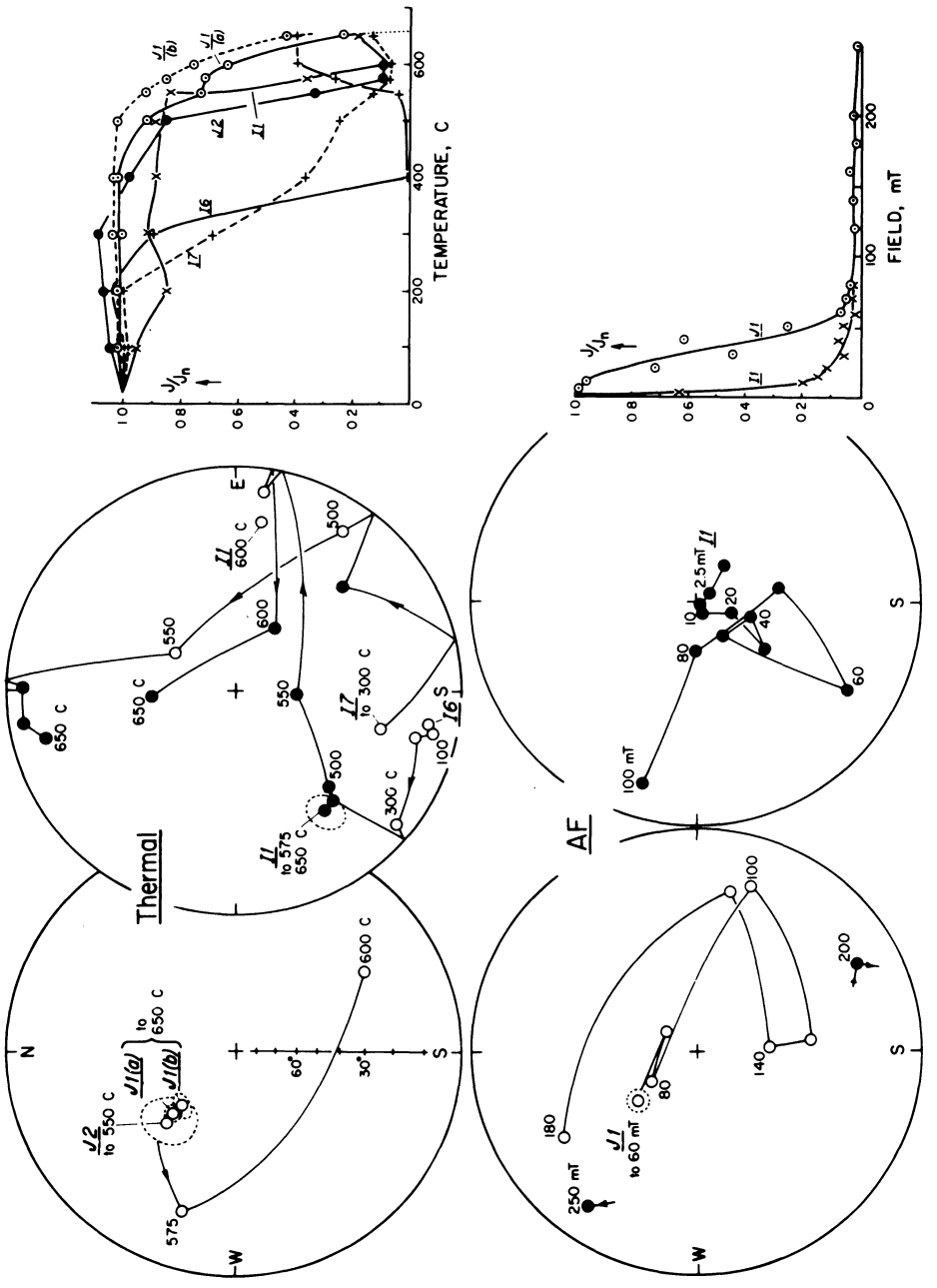


Fig. 2. Pilot thermal and alternating-field (AF) demagnetization of selected andesitic specimens from the sites shown (underlined). Specimens used in comparing thermal and AF treatments were taken from different samples at a given site (J1, I1); also specimens J1(a) and J1(b) are from different samples. Open circles indicate N. pole directions upward; filled circles, downward. Within dashed outlines, directions remain stable up to the highest step shown. The stereograms are Wulff nets.

3.1 Natural remanence (NRM) and related properties

Most of the sampling sites (Fig. 1) are weakly magnetic, though at some sites the NRM intensity (J_n) varies widely. J_n values for 60 samples range from 0.1–0.6 ($\times 10^{-3}$ A/m) for the acid tuffs (sites C1–C3) to 2,000 ($\times 10^{-3}$ A/m) for the exceptionally magnetic "andesitic tuffs" (site J1), and at 17 sites of andesites (sills, pillows, flows) the J_n distribution is 0.2–50 ($\times 10^{-3}$ A/m). The intensities of the rhyolites (site D), mafic intrusives (sites A1, E3) and "acid-intermediate tuffs" (sites E1, E2) also span this broad range. Such variable but relatively low intensities are typical in rocks of this age and composition, making it cumbersome to measure the weakest remanences accurately. Five tuff sites (A2, C1–C4) are omitted from Table 1 due to this cause, but the NRMs of five specimens from sites C1, C2 and C3, treated as one site, were measured on the cryogenic magnetometer at Ohio State University through the courtesy of Prof. H. Noltimier (Table 1).

Initial volume susceptibility (K) was measured on 31 samples (one specimen per sample) of andesite sills (sites I1–I8, J2), pillows (site G) and tuff (site J1). All but one specimen from the 10 sill and pillow sites gave low susceptibilities in the range 0.1–1.1 ($\times 10^{-3}$ SI units), while the "magnetic" site J1 gave values of 2.5–50 ($\times 10^{-3}$ SI units). Koenigsberger ratios $Q_n (= J_n/KH$, where $H = 40$ A/m is the present field) were calculated for these 31 specimens. Only one Q_n value exceeds unity (2.3 at site G), the other 30 being in the range 0.01 to 1. Such findings are not uncommon in older, e.g. Paleozoic, volcanic rocks and indicate that induced and low-stability components dominate the NRM. While this is partially confirmed in Fig. 2, the low Q_n ratios have not precluded the extraction of a stable remanence through cleaning, usually at intermediate fields or temperatures (Figs. 2 and 5; Table 1). Conversely, the presence of a stable remanence and relatively high Q_n in

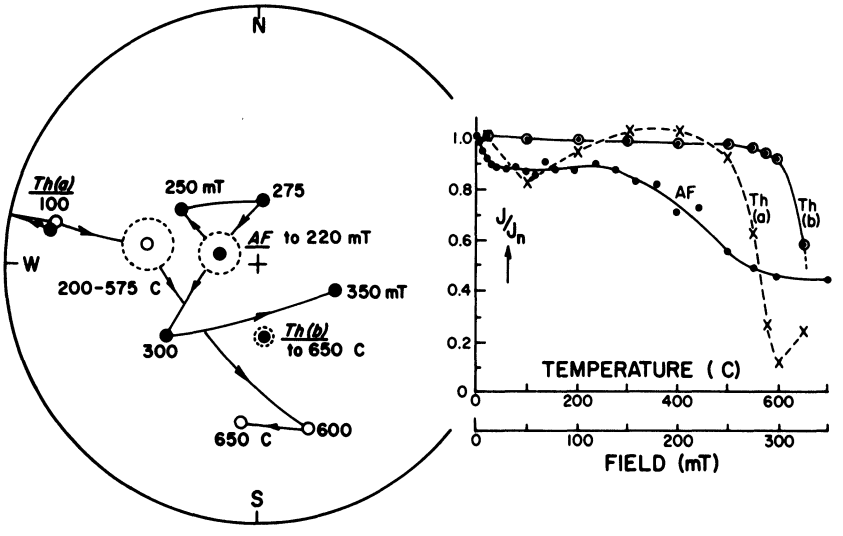


Fig. 3. Pilot thermal and AF demagnetization of selected pillow andesite specimens from the "anomalous" site G. The three specimens are from different samples. Notations are in Fig. 2. Note how sharply these stable remanence directions of different specimens diverge.

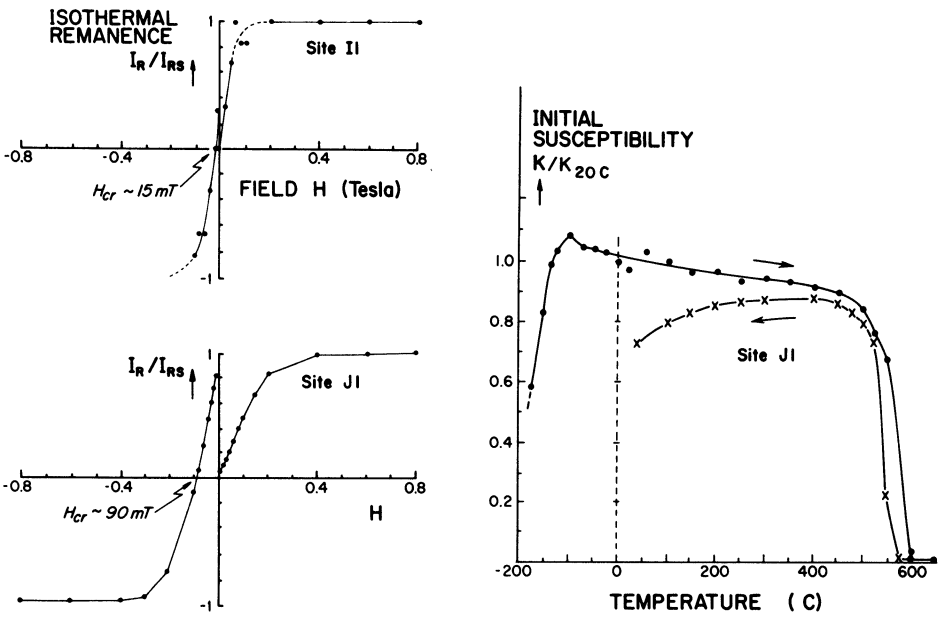


Fig. 4. Other magnetic properties, for three specimens from sites I1, J1. *Left.* The normalized isothermal remanence is shown, where I_R is the remanence acquired stepwise in fields H, I_{RS} the saturation remanence and H_{cr} the coercivity of remanence. The I1 specimen is typically soft, the J1 specimen typically hard. *Right.* Temperature dependence of magnetic susceptibility (K) in low field (0.03 mT), normalized, for a specimen heated and cooled in air. The heating curve indicates multi-domain magnetite (see text), which may depart from stoichiometry since the peak occurs at a temperature above the isotropic point (−155 C) for magnetite.

a suite of rocks does not ensure that their directions are significantly grouped, a case in point being site G (Fig. 3).

Nine of the 24 site-mean NRM directions in Table 1 are random at the 95% confidence level (IRVING, 1964, p. 63), and the precision (k) of the others is generally not high. Figure 5 (upper left) shows that the between-site grouping also is poor.

3.2 Pilot demagnetizations

One specimen each of 15 samples from 11 sites (B1–3, D, E1–3, F1, F2, H1, H2) were given AF treatment in 12 steps to 80 mT. The early steps showed a trend towards steeper negative remanence directions to northwest, compared with the more horizontal NRMs. This trend can be seen best in the site-mean directions (Fig. 5, top). The smallest vector movements and tightest between-sample groupings generally occurred in the AF range 15 to 40 mT, while at higher fields the directions became increasingly randomized. Between the NRM and the 40 mT step the intensity J_n decayed by an order of magnitude in some specimens, but less in magnetically harder specimens.

One specimen each of 29 samples from 11 other sites (I1–8, G, J1, J2) were thermally demagnetized in 9 steps to 650 C. As was found for the AF-treated sites, the demagnetized remanence vectors tend to point upward and to the northwest in the case of

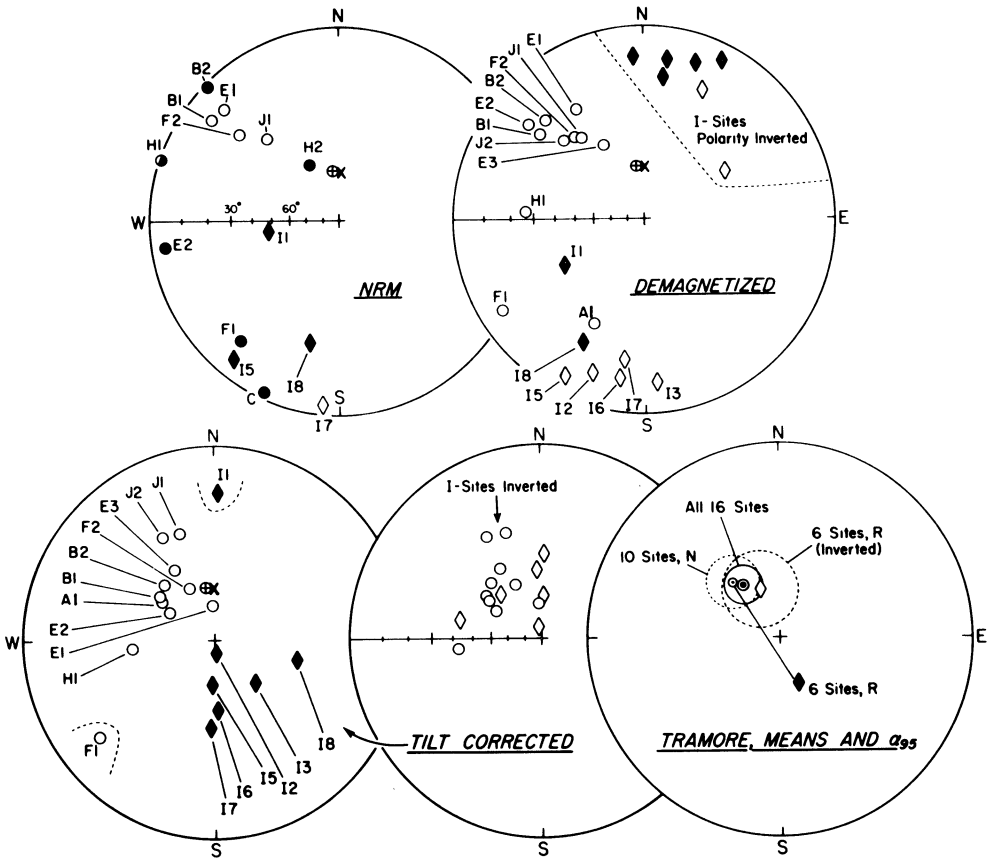


Fig. 5. Site-mean and (bottom right) overall mean remanence directions, excluding sites having random within-site grouping. See also Tables 1, 2. The NRM directions, demagnetized directions (AF or thermal) and demagnetized plus tilt-corrected directions are shown. Diamonds denote I-sites; circles, all other sites. Open symbols indicate N. pole directions upward; filled symbols, downward. X and circled cross denote the axial dipole and present field directions (downward) at Tramore. Dashed lines separating inverted I-sites (top right) and two aberrant sites (bottom left) have no statistical significance. *Bottom right.* Circled dot and open diamond denote group mean directions for the normal (N) and inverted reversed (R) sites, with 95% confidence (α_{95}) circles shown dashed. The N and R mean directions have been joined to depict the slight antiparallelism. The circled thick dot is the Tramore mean direction, shown with solid α_{95} circle. Wulff net.

sites J1 and J2, but for sites I2 through I8 the directions are shallow and southerly (Fig. 5, top), for a reason that becomes clear later. Smallest vector movements and best groupings occurred usually below 300 C, and some directions began to be scattered just above 300 C. Six examples (Fig. 2, top) from moderately to highly stable specimens illustrate these trends. With one exception (I7) the thermal decay curves show mainly single discrete blocking ranges near 400, 600 or 650 C, and these may correspond to a primary component. The reason for the intensity rise at high temperature in three of the curves is

not clear, but similar observations have been reported from thermal demagnetization by other workers and may be due to viscous remanence acquisition. Figure 2 (top, middle) suggests that systematic high-temperature components are not the cause.

Six fresh specimens from sites I1, I2, I6, G and J1 were demagnetized in alternating fields between 100 and 350 mT maximum to compare with 13 thermally demagnetized specimens from the same sites. Examples (Fig. 2) show that agreement between the two treatments is excellent in the case of the very stable J1 specimens. The agreement is only moderate for the specimens from site I1 used in the figure, which are representative of less stable behavior at the nine andesite sill sites (compare AF and thermal decay curves for site I1). In the case of site G, three examples (Fig. 3) illustrate the peculiar results produced by both treatments: whereas the remanence is, on the whole, very hard, the directions observed in different specimens are grossly discordant, irrespective of the treatment used. Not surprisingly, site G did not yield a meaningful remanence (Table 1). At the other sites being compared (I1, I2, I6, J1), the thermal treatment tended to produce more consistent results than the AF treatment, while the AF results in turn were more erratic than those of the AF pilot demagnetizations on sites B1–3, E1–3, F1, F2, H1 and H2. Ultimately, thermal demagnetization was found preferable at sites I1–I8, G, J1 and J2, and AF treatment at the other sites.

3.3 General treatment

Accordingly, the 23 sites (omitting C1–C3) were treated as follows: one specimen from every sample was demagnetized by AF (12 sites) or thermally (11 sites) in two to four steps. Table 1 shows the treatments adopted, based on the criterion of best vector grouping.

Four of the 23 demagnetized sites (I2, I3, I5, I7) had initially very poor precision due to the presence of one grossly discordant specimen each, and these 4 specimens were omitted (Table 1) to improve precision, a step that is here justified by the fact that it left the overall mean direction for these four sites unchanged. At 4 other sites (B3, D, G, I4) the within-site vector groupings after demagnetization are random, and these sites were excluded from the final statistical treatment. One high-precision site (H2) was excluded as well, due to uncertainty in the geologic structure at that site, which was not realized until this work was completed (W. Stillman, personal communication). This leaves 18 significant site-mean directions, 14 of them with improved precision due to cleaning.

Demagnetization has resulted in two clearly distinct site groupings (Fig. 5, top right) having remanence directions to the south (mainly I-sites) and northwest, each group showing increased between-site precision compared to the NRM. It was suspected that the two groups might be of opposite polarity, but this cannot be the sole cause of the misalignment which is roughly 90°. The I-sites have been plotted also with inverted polarity to illustrate this fact.

3.4 Remanence stability and origin

Demagnetization has typically removed a component whose mean direction is between the NRM and the present (downward) field direction. This applies to sites in both of the directional groupings and can be verified by inspection of Fig. 5 (top). Exceptions

are highly stable specimens (e.g. Fig. 2, left diagrams) that seem to lack a secondary component. At all sites except H2, the site-mean directions both before and after demagnetization depart from the present and axial dipole directions by large angles (Fig. 5, top). These facts, along with the pilot treatment results, support the supposition that the remanence isolated by demagnetization has long-term stability and represents a geologically ancient Earth's field direction.

In the andesite sills (sites I1–8, J2) magnetite or titanomagnetite is probably the only remanence carrier, as is suggested in Fig. 2 by the generally simple trends of the thermal decay curves showing that the magnetization is blocked below the Curie temperature (T_c) of magnetite, and by the low value (<10 mT) of the median demagnetizing field ($H_{1/2}$) for site I1. Although this particular value reflects an untypically low coercivity (H_c), the more typical finding of medium-hard components ($H_c \sim 10$–20 mT) in the andesite sills and also at most other sites is likewise consistent with (titano)magnetite as the sole remanence carrier. This interpretation is supported by a test of isothermal remanence (IRM) using a fresh specimen from site I1 (Fig. 4). High-field IRM is diagnostic of the entire magnetic fraction and shows that this material is saturated below 100 mT and has a low coercivity of remanence ($H_{cr} \sim 15$ mT).

For the exceptionally stable site J1, the decay curves show a hard component ($H_{1/2} \sim 40$ mT) and a blocking range extending to temperatures above 600 C. Site G shows similar properties. IRM curves (Fig. 4) show that saturation is not reached until 400 mT and that H_{cr} is high (~ 90 mT) suggesting that the magnetic carrier at site J1 (and site G?) may be highly cation-deficient magnetite or perhaps magnetite plus hematite. Finally, a test of susceptibility vs temperature (K-T, Fig. 4) yielded a high T_c (600 C) and a low-temperature peak on heating. Such peaks are diagnostic of multi-domain (MD) magnetite (RADHAKRISHNAMURTY and DEUTSCH, 1974). Therefore this specimen contains more than one component, but since MD magnetite is unlikely to contribute to stable remanence, the remanence is probably carried by cation-deficient single-domain magnetite or by hematite or both.

4. The Mid-Ordovician Field in SE Ireland

After correction for geologic tilt the site-mean directions become further polarized in two nearly antiparallel groupings (Fig. 5, bottom left) comprising respectively the I-sites and all other sites. One site-mean direction in each group (F1, I1) is sharply misaligned with the remaining directions in that group and these two sites were omitted from final averaging to improve precision (Table 1). The overall mean direction (Fig. 5, bottom right) is nearly unaffected by this step. Group mean directions for the 16 remaining sites (10 and 6 sites each) are given in Table 2, showing that the tilt correction has moderately improved the between-site precision for each group. This could indicate that the remanence isolated by demagnetization preceeded tectonic deformation.

4.1 Significance of the site groupings

To obtain the mean vector for all 16 sites the two groups can be combined in four ways. The 168° difference between the two tilt-corrected group-mean directions suggests

Table 2. Tramore area: overall mean remanence directions and Fisher statistics, after demagnetization.[1]

		Before tilt correction					After tilt correction				
	N	$D(°)$	$I(°)$	R	k	$\alpha_{95}(°)$	D (°)	I (°)	R	k	α_{95}(°)
All except I-sites[2]	10	307	−43	8.957	8.6	17	319	−59	9.594	22	11
I-sites only[3]	6	195	−10	5.629	14	19	156	+68	5.718	18	16
All sites (I-sites unchanged)	16	258	−46	9.840	2.4	31	307	−46	4.139	<2	69
All sites (I-sites inverted)[3]	16	337	−27	11.270	3.2	25	*324.3*	*−62.6*	*15.243*	*20*	*8.5*

Tramore: present field direction, 349°, +68°; axial dipole direction: 0°, +69°
Paleomagnetic N pole position:[4] 11°S, 162°W ($dp=10°$, $dm=13°$, $N=16$ sites)

[1] N=number of sites; R=resultant of N unit vectors; other notations as in Table 1.
[2] Normal polarity.
[3] Reversed polarity indicated in Table 3; hence polarity of I-site directions is inverted for final averaging (last row).
[4] Calculated from tilt-corrected data in last row (*italics*).

Table 3. Tramore area: F-ratio test of alternative directional groupings.

Alternative[1]	Overall mean direction based on		G	N	δ_b^2	δ_w^2	δ_b^2/δ_w^2	$F_{2,28,0.01}$	$F_{2,28,0.05}$
	Tilt correction (all sites)	Inverted polarity (I-sites)							
1	no	no	2	16	9.49	0.202	47	5.45	3.34
2	no	yes	2	16	6.63	0.202	33	5.45	3.34
3	yes	no	2	16	22.3	0.098	227	5.45	3.34
4	yes	yes	2	16	0.138	0.098	1.40	5.45	3.34

[1] See also Table 2. G and N=numbers of site groupings and sites. δ_b^2 and δ_w^2 (in radians^2)=between-group and within-group angular variance. Last two columns=F-ratio, $F_{2(G-1),2(N-G),\alpha}$, at 1% and 5% probability levels.

that the respective sites have opposite polarity and that the site-mean directions of one group should be inverted for calculating an overall mean direction. Here one makes two basic assumptions: the sites were magnetized in fields of respectively normal and reversed polarity, and the field of either polarity was that of a centered axial dipole. However, since all I-sites have been tilted through 90 degrees or so, it is not immediately obvious that dual polarity combined with tectonic deformation is the statistically favoured mechanism. Other alternative are: 1) The stable remanence post-dates the tilting and there was no reversal. 2) Remanence post-dates tilting, and reversal occurred. 3) Remanence pre-dates tilting, and no reversal. Fisher statistics for these and the fourth alternative (remanence pre-dates tilting and reversal occurred) are given in Table 2. As is clear also from Fig. 5, alternative 3 corresponds to the lowest precision, but the k-values of alternatives 1 and 2 also are low.

The 4 alternatives were then submitted to an F-ratio test (WATSON, 1956; LAROCHELLE, 1967). To apply this, one must assume that the variances of the two populations (i.e., site groupings) being compared are approximately equal. This assumption is reasonable since in each of the four cases tested the ratio of appropriate k-values (Table 2, first two lines) is 2 or less. Table 3 shows that, for alternative 4, the I-sites are not distinct from the remaining sites at 5% probability level, whereas in each of the three other cases it is more than 99% probable that the two site groupings are distinct. It was concluded that the most probable explanation of the paleomagnetic results is alternative 4, according to which the stable remanence (i) was acquired prior to deformation of the Tramore volcanics and (ii) carries a record of opposite polarities of the Earth's magnetic field.

4.2 Fold test

Conclusion (i) represents a positive fold test with a sharply improved precision k (from 3.2 to 20, Table 2). Both the fold test and the presence of both polarities constitute additional evidence for long-term stability of the remanence. The fold test shows that this remanence pre-dates the Caledonian orogeny and, considering the stability evidence already discussed, it is reasonable to conclude that the remanence was in fact acquired at

about the time the rocks were formed in the mid-Ordovician.

4.3 Field reversal and paleomagnetic pole

The mean direction of the larger group of sites (319°, −59°; Table 2) corresponds to a north pole located near the apparent polar wandering path for stable western Europe and so was taken to be of normal (N) polarity; then the second group (I-sites) are reversed (R). Since the I-sites are older, a field reversal from R to N polarity must have occurred sometime between the respective times these sites and the other westerly sites (Fig. 1) became magnetized, i.e. most probably during the Llanvirn or early Llandovery.

The overall mean direction obtained by combining normal and inverted reversed sites is 324.3°, −62.6° ($\alpha_{95}=8.5°$). Assuming an axial dipole field, this corresponds to a geographical north pole in the present Pacific Ocean, at 11°S, 162°W ($dp=10°$, $dm=13°$, $N=16$ sites).

Since at most sites the within-site vector grouping is comparable or even inferior to the between-site grouping, it is difficult to assess any contribution from secular variation to the size of the error oval, and this will not be attempted. A second possible source of scatter is apparent polar wandering within the time span from the oldest (R) to the youngest (N) sites. This is also difficult to estimate, since the 12° departure from antiparallelism of the R and N group mean directions is not significant and neither is the separation between R and N south poles in Fig. 6 (Table 2). For the same reason, however, the magnitude of any drift of SE Ireland or of polar wandering cannot have been large during the age span of Tramore sites.

5. Comparison with Other Paleomagnetic Data

The Tramore pole (Fig. 6, T) deviates by 30 degrees from a pole (M) of reversed polarity based on the mid-Ordovician Mweelrea ignimbrites (MORRIS *et al.*, 1973) which

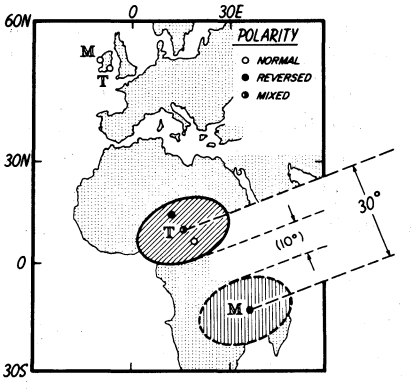


Fig. 6. South pole positions for the Tramore volcanics (T, this study) and Mweelrea ignimbrites (M, MORRIS *et al.*, 1973) shown with 95% confidence ovals. Pole T is at 11°N., 18°E (dp, $dm=10°$, 13°), pole M at 11°S, 38°E (dp, $dm=10°$, 18°). Poles are shown separately, without ovals, for the Tramore normal sites (8°N, 21°E; dp, $dm=12°$, 16°) and reversed sites (16°N, 13°E; dp, $dm=23°$, 28°). Plotted on the present geography.

lie across the proposed plate suture from Tramore. Like the Tramore volcanics, the ignimbrites yield a positive fold test (DEUTSCH and SOMAYAJULU, 1970; MURTHY and DEUTSCH, 1971; MORRIS *et al.*,1973) and the M pole is believed to represent the Llanvirn age of the ignimbrites. As Llanvirn is also the favoured age of the R sites at Tramore, the Mweelrea rocks conceivably became magnetized during the same reversed polarity epoch as the Tramore R sites. Since the R sites predate the N sites, some of which are post-Llanvirn, the Tramore rocks (N and R sites) are *on average* probably somewhat ($\leqq 10$ Myr) younger than the Mweelrea sites.

The M and T pole positions are significantly different and can be reconciled only through relative rotation between the two sampling localities. A notable feature of Fig. 6 is the fact that localities M and T lie approximately on a common great circle with the T poles (reversed, combined, normal) and the M pole, in that order. The great circle is a paleomeridian and one sees that, *going back in time*, any simple plate rotation designed to make the M and T poles coincide must involve an increasing separation between the M and T sampling localities roughly along the present NW-SE trend (i.e., ancient N-S trend) of the alignment. This trend is nearly perpendicular to the strike of the plate suture (62°: PHILLIPS *et al.*, 1976). Such a reconstruction is consistent with the hypothesis that the two localities, which are now 2° apart, lay on opposite (north and south) sides of a mid-Ordovician Proto-Atlantic ocean. From the paleolatitudes of Mweelrea ($\lambda_p = 14°S$) and Tramore ($\lambda_p = 44°S$) this ocean at least partly occupied the southern hemisphere and was at least 28° wide, not considering paleomagnetic error.

I had argued earlier (DEUTSCH, 1969) that a large discrepancy between the then available Mweelrea pole (DEUTSCH and SOMAYAJULU, 1970) and British Ordovician poles was consistent with the existence of a possibly wide Proto-Atlantic ocean in this section of the Caledonides. While further work, (MURTHY and DEUTSCH, 1971; MORRIS *et al.*, 1973) left the Mweelrea pole essentially unchanged, it was counter-argued (BRIDEN *et al.*, 1973; MORRIS, 1976) that the aberrant magnetic direction of the ignimbrites relative to British Ordovician rocks is due to local tectonics in NW Ireland rather than an intervening ocean. Specifically, MORRIS (1976) reasoned that development of the South Mayo trough during the Devonian caused the ignimbrites to rotate first 60° anticlockwise, then 30° clockwise, producing the observed southeasterly (130°–135°) declination; one assumes here that the original direction was parallel to the south or north ($\sim 170°$, $\sim 350°$) declinations characteristic of British Ordovician localities. As a further argument against a substantial Proto-Atlantic, BRIDEN *et al.*, (1973) pointed out that the small difference between the observed magnetization directions north of the hypothetical suture in Britain (Aberdeenshire gabbros: SALLOMY and PIPER, 1973) and south of the suture is not significant.

These were persuasive objections in the light of then available paleomagnetic data. The results presented here override the objections and require reconsideration of the case for a wide Proto-Atlantic. In Fig. 7, the results of nine paleomagnetic studies of British and Irish Ordovician rocks are plotted as south poles. Only those studies are included in which it is reasonably sure that the magnetization was acquired within the Llanvirn-Llandeilo-Caradoc time span (473–440 Myr). Several reliable but anomalous mid- to late Ordovician poles in high latitudes (THOMAS and BRIDEN, 1976; PIPER, 1979) have been

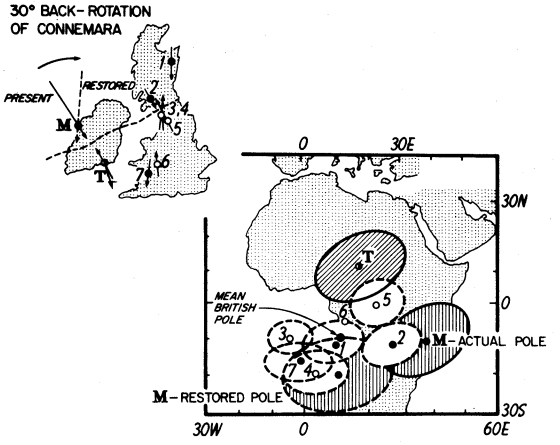


Fig. 7. Paleomagnetic south poles from mid- to late Ordovician (Llanvirn to Caradoc) igneous rocks in the British Isles, plotted on the present map. The Irish poles (T, M) are as in Fig. 6, pole M being shown also in a "restored" position obtained by reversing a postulated 30° net anticlockwise tectonic rotation of the region bearing locality M. Open, filled and divided circles denote normal, reversed and mixed polarity. Numbers in square brackets below are declinations of the remanence vector shown by arrows on the location map. *British poles*, 1. Aberdeenshire gabbros, Scotland [168°] (SALLOMY and PIPER, 1973). 2. Ballantrae gabbros and serpentinites, Scotland [148°] (PIPER, 1978). 3. Eycott Group lavas, England [0°] (BRIDEN and MORRIS, 1973). 4. Carrock Fell gabbros-granophyres, England [350°] (BRIDEN and MORRIS, 1973). 5. Borrowdale volcanics, England [329°] (FALLER *et al.*, 1977). 6. Breidden Hill intrusive inlier, Wales [341°] (PIPER and STEARN, 1975); for clarity, the large confidence oval (*dp*, *dm*=15°, 22°) has been omitted. 7. Builth volcanics, Wales [177°] (PIPER and BRIDEN, 1973). The mean position of poles 1–7 (large filled circle) is 10°S, 10°E. *Irish poles*. M. Mweelrea ignimbrites [131°] (MORRIS *et al.*, 1973). T. Tramore volcanics [N sites, 319°; R sites, 156°; combined sites (R inverted), 324°] (This study). The mid-Ordovician paleolatitudes of Mweelrea (M) and Tramore (T′) (Fig. 8) are 14°S and 44°S, respectively.

excluded as it seems that these results cannot be explained by regional tectonics (PIPER, 1979) and so are not directly relevant here.

Five of the seven British poles (Fig. 7, nos. 1, 3, 4, 6, 7) were obtained from rocks with average north-south magnetization. When the Mweelrea ignimbrites are rotated 30 degrees clockwise so as to reverse the inferred net deformation of Connemara, pole M becomes statistically coincident with these British poles, apparently conforming to the argument of BRIDEN *et al.* (1973) and MORRIS (1976). However, the local restoration of Connemara not only fails to reduce the discrepancy between poles M and T, but makes it worse. This follows directly from the 30° paleolatitude difference and the shared paleolongitude of localities M and T, showing that no amount of horizontal rotation at either locality will reconcile the two poles. Moreover, two poles reported since 1976 (nos. 5, 2) are associated with mean NW-SE declinations and are each distinct from British poles 1, 3, 4 and 7: Pole 5 (Borrowdale volcanics) is an improved result from the English Lake District (previous declination=0; BRIDEN and MORRIS, 1973) south of the plate suture in Britain, and pole 2 is derived from the Southern Uplands Blocks, north of the suture. These two pole positions are statistically identical to Irish poles T and M,

respectively. Then two poles (M and 2) based on rocks north of the suture diverge from the Tramore pole and, marginally, from the Borrowdale pole, both representing rocks south of the suture. As each of these four poles (M, 2, T, 5) corresponds to a NW-SE directed magnetization of similar age, it is difficult to maintain an anomalous status for pole M.

Thus, while tectonic flexuring in NW Ireland cannot be excluded as a cause of the remanent declination of the Mweelrea ignimbrites, the deformation responsible would have to have affected a much larger region of Ireland and Britain on both sides of the suture to explain plausibly why the (NW-SE) declinations at four localities conform. It seems highly improbable that the concordant vector orientations at these localities resulted from separate local block rotations about vertical axes.

It remains to account for the presence of two sets of Ordovician exposures in the British Isles, each set with characteristic declinations (N-S, NW-SE). It seems clear that this requires a change in direction of the Earth's magnetic field relative to the exposures, occurring *during* the Ordovician. Alternative mechanisms could be 1) plate motion implying, however, that *two* plates that were separated by an ocean shared a rotation in the same sense; 2) polar wandering; or 3) an excursion of the geomagnetic field. Further paleomagnetic work is needed to solve this problem.

6. Width of the Proto-Atlantic Ocean

The postulate of an open Proto-Atlantic ocean in the Ordovician will now be further considered with respect to Ireland only. The paleomagnetic data have demonstrated the *existence* of such an ocean, but its *size* (other than *minimum* size, from paleolatitude differences) is more elusive. The ancient separation of a pair of sampling localities now on opposite sides of a suture (e.g. Tramore and Mweelrea) cannot be determined from the two pole positions unless the pivot (rotation pole) is known, which is seldom the case. The pivot position can be determined, however, if the directions of approach of the two plates effecting ocean closure are known or may be assumed.

For this purpose I have taken the Caledonian plate tectonic model of PHILLIPS *et al.*, (1976) as a working hypothesis in testing the paleomagnetic results. This model represents an attempt to reconstruct in detail the Ordovician to Devonian continental drift history of the British Isles, and it includes quantitative estimates of the geometry of ocean closure. Although Phillips *et al.* estimated also the minimum Proto-Atlantic width (600–800 km in the Arenig) from calculated closure rates, an estimate of the actual ocean width exclusively from their model would involve much extrapolation, which is not the case when paleomagnetic poles are being compared. The main relevant features of the model are, briefly: a non-parallel approach of two plates ("NW plate" and "SE plate") and their associated subduction systems in the Ordovician, resulting in oblique collision from late Ordovician to early Devonian time and subsequent (Devonian) dextral strike-slip movement by 1,270 km along the ENE-WSW trending suture. The directions of approach relative to the present geography are 155° (NW plate) and 305° (SE plate), allowing for 50% post-collision strain.

I assumed for convenience that the SE plate moved alone relative to a "fixed" NW

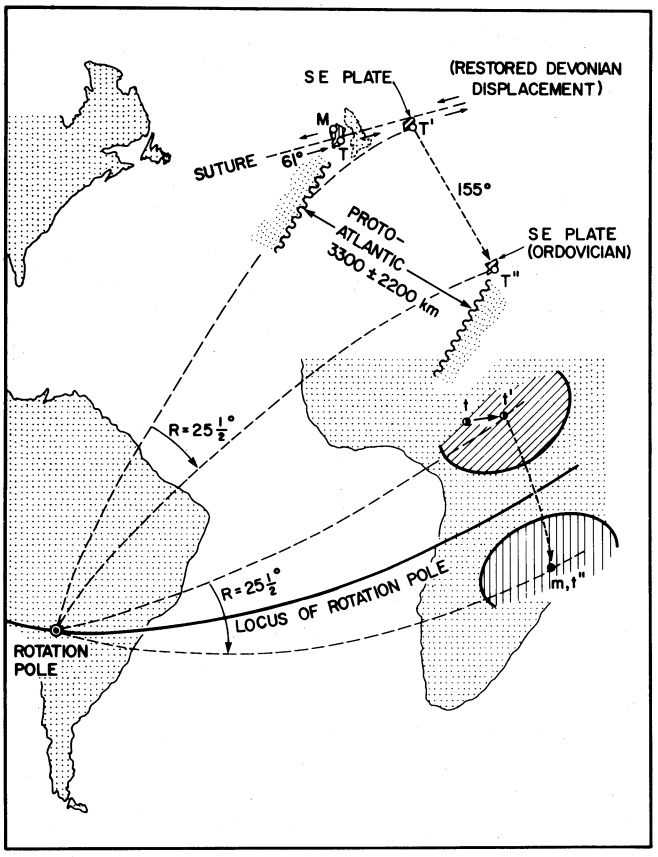


Fig. 8. Proto-Atlantic reconstruction relative to the Irish Caledonides, plotted on the present map (see text). Continental drift patterns inferred by PHILLIPS *et al.* (1976) have been combined with the paleomagnetic results to yield a rotation pole (27°S, 67 $^1/_2$°W) for Proto-Atlantic closure in post-mid-Ordovician time. Poles t, m correspond to the Tramore and Mweelrea poles (T, M) in Fig. 6, and t′ (12°N, 26°E) is the new position of pole t which corresponds to the restored position T′ (56 $^1/_2$°N, 11°E) of Tramore when the inferred Devonian strike-slip movement of SE Ireland is reversed. The rotation pole locus is equidistant from poles t′ and m. Rotation, backwards in time, of the SE Plate from T′ to its "final" (i.e. Ordovician) location T″ (34 $^1/_2$°N, 27°E) about the rotation pole corresponds to rotation of pole t′ into coincidence with pole m (=t″). The dashed lines from the rotation pole to points T′ and T″ may be visualized as plate lever arms intersecting the drift path T′ T″ at a right angle. The limits (±2,200 km) quoted for the Proto-Atlantic width estimate correspond to the combined 20° paleomagnetic error in the two confidence ovals.

plate, approaching it in a 320° direction which is the resultant of the separate NW and SE plate motions. For reconstructing the drift geometry one reverses the above sequence (Fig. 8), starting with a 1,270 km eastward displacement of the SE plate along the suture that carries the Tramore volcanics to an "early Devonian" location (T′). Going further back in time, one "opens" the ocean by moving the SE plate in an intial 155° direction from site T′ (obtained by reversing the 320° approach direction and adding 15° for

convergence of meridians between sites T and T′). A single-step motion is assumed. To be found is the original (Ordovician) position T″ of the SE plate which results when the displaced Tramore pole t′ is made to coincide with the Mweelrea pole (m). The unknown angular displacement $p(=\mathrm{T'T''})$ is given by (DEUTSCH, 1969):

$$1-\cos p=\sin^2\theta\ (1-\cos R) \tag{1}$$

$$1-\cos p'=\sin^2\theta'\ (1-\cos R) \tag{2}$$

where p' is the (known) displacement t′m $(=\mathrm{t't''})$ between paleomagnetic poles, θ and θ' are the distances from the rotation pole to T′ and t′ respectively, and R is the angle through which the plate (T′) and the pole (t′) are rotated. Then the rotation pole (or its antipole) lies at one intersection of the rotation pole locus with a great circle that is drawn orthogonally to the drift path T″T′ at point T′. The rotation pole thus obtained (27°S, 67½ °W) determines θ and θ', yielding R (25½°) and p (24½°) in Eqs. (1) and (2).

From p one finds the pre-closure position T″ of the SE plate and so the width TT″ of the inferred Proto-Atlantic ocean relative to the Irish Caledonides. The value obtained (Fig. 8) is $30°\pm20°$ (or $3{,}300\pm2{,}200$ km), where the limits correspond to paleomagnetic error using the 10° least distance between the two confidence ovals as a rough criterion for the minimum width of the ancient ocean. The 3,300 km value is comparable with the present width of the North Atlantic, showing that the paleomagnetic results reported here are compatible with a wide Proto-Atlantic ocean in mid-Ordovician time.

After the reconstruction of Fig. 8 had been made, PHILLIPS (1980) proposed a revision of the PHILLIPS *et al.*, (1976) model, in which the postulated angle of approach of the SE plate and magnitude of the subsequent dextral strike-slip movement are changed to 295° and 600 km, respectively. It can be shown, however, that substitution of these figures makes very little difference to the value of Proto-Atlantic width calculated from the older model. In either case a reduction of the large confidence limits associated with that value will be possible when Ordovician paleomagnetic data having less error become available.

6.1 Comparisons across the North Atlantic

In making the above reconstruction (Fig. 8) the present geographical coordinates have been used as a convenient reference. Allowance for the post-Paleozoic opening of the Atlantic (BULLARD *et al.*, 1965) must be made, however, if Irish or British Lower Paleozoic paleomagnetic results are to be compared with North American results of the same age. Such comparisons are a necessary step towards reconstructing the total Proto-Atlantic paleogeography which is now poorly understood. For example, the two Irish Ordovician pole positions (Tramore, Mweelrea), after correction for the Bullard fit, still deviate considerably from representative mid- to late Ordovocian poles for cratonic North America (MCELHINNY and OPDYKE, 1973; VAN DER VOO and FRENCH, 1977). Also, early Ordovician poles for eastern and western Newfoundland are far apart (DEUTSCH and RAO, 1977), compatible with a substantial Proto-Atlantic ocean whose size, however, cannot be quantified on the basis of existing Newfoundland data. These problems will be discussed in a subsequent paper.

It is a pleasure to acknowledge the following help received. Logistical support for field trips was generously provided in 1971 by Dr. C. E. Williams, Director, Geological Survey of Ireland and in 1976 by Prof. Thomas Murphy, Director, Dublin Institute of Advanced Studies. Dr. Elmar Schiener, University College, Dublin (1971) and Dr. Chris Stillman Trinity College, Dublin (1971, 1976), gave detailed geological information in the field and later on. David Waddell (G. S. I.) and Dr. David Howard (D.I.A.S.) assisted the author in the field. Maeve Boland, Dr. Adrian Phillips (Trinity College) and others provided crucial discussion. Jim Sharpe and David Morgan did most of the laboratory measurements at Memorial University of Newfoundland. Prof. Hal Noltimier, Ohio State University, made his cryogenic magnetometer available for measuring some weak specimens. This research was supported by Grant A-1946 from the Natural Science and Engineering Research Council of Canada.

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